Autodesk RAG Chatbot

Senior ML Engineer Take-Home Assessment

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Ingestion	Retrieval	Generation	Evaluation
Multi-layer HTML parsing + chunking	Hybrid search + Sentence Window	Grounded prompting + hallucination guard	TruLens RAG Triad 10 experiments

Problem Statement & System Architecture

What was built and why

The Challenge

- 1,218 noisy HTML files — JS shells, marketing pages, help docs, e-commerce
- Two pipeline modes: corpus-only (Task 1a) and blended web search (Task 1b)
- Evaluate quality with a principled, research-backed methodology
- All open-source, fully local — no data leaves the machine

System Flow

01 **HTML Corpus**
1,218 pages → parse → chunk → embed

02 **Hybrid Retrieval**
Vector + BM25 → RRF → Rerank

03 **LLM Generation**
gemma3:4b (Metal GPU) + grounded prompt

04 **Two UIs**
Streamlit App 1 (8501) + TruLens App 2 (8502)

05 **Evaluation**
TruLens RAG Triad — 10 experiments tracked

Ingestion — Data Challenge & Preprocessing

Multi-layer extraction + product normalization

1,218	~19%	~80	3
HTML Files	Filtered Out	Product Aliases	Extraction Layers

Multi-Layer Extraction Pipeline

- Layer 1 — trafilatura (favor_recall=True): keeps competitor comparisons, feature sidebars
- Layer 2 — BeautifulSoup on COPY of original DOM: v1 bug ran BS4 on already-stripped soup
- Take the LONGER result — competitor names (SolidWorks), version strings survive
- Layer 3 — html2text fallback if both < 100 chars

Document Selection & Normalization

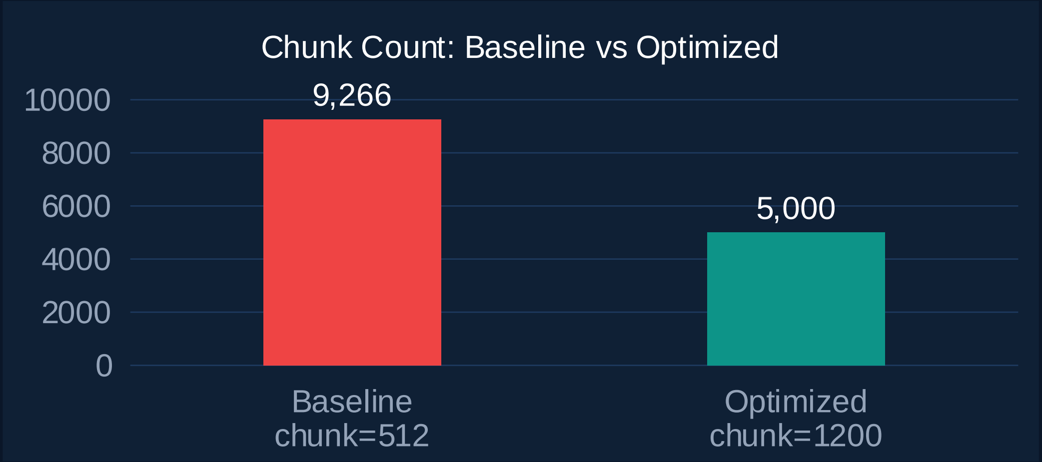
- Ingest ALL docs passing quality gate (>100 chars) — coverage > precision
- Product alias table: ACDIST→AutoCAD, F360→Fusion 360, ACAD_E→AutoCAD Electrical
- Without aliases: BM25 cannot match 'F360' against a user query for 'Fusion 360'
- Tables → Markdown before stripping: keeps comparison tables intact for LLM

Ingestion — Data-Driven Chunking Strategy

Corpus analysis drove the chunk size decision

Corpus Analysis

Median doc length	1,235 chars
P75	2,666 chars
P90	5,877 chars
Baseline chunks	9,266 (chunk=512)
Optimized chunks	~5,000 (chunk=1200)
Docs kept whole	~50% at chunk=1200



💡 Why 1,200 chars?

At chunk=1200, ~50% of documents stay as a single chunk — the LLM sees the full page in one retrieval hit. Smaller chunks fragment coherent answers and flood the index with noise.

Retrieval — Hybrid Search Pipeline

Vector + BM25 → RRF Fusion → Cross-Encoder Reranking

1. **Vector Search**
   BGE-small-en-v1.5
   +ChromaDB
   top-15 candidates

2. **BM25 Search**
   Bigram tokenizer
   exact product names
   top-15 candidates

3. **RRF Fusion**
   BM25=0.4
   Vector=0.6
   no score normalization

4. **Cross-Encoder**
   ms-marco-MiniLM
   joint (query, passage)
   reranking

5. **Top-5 → LLM**
   grounded prompt
   +citation instructions
   +chat history

Why Vector alone fails on Autodesk queries

"AutoCAD LT 2024" — the embedding model maps this to a dense vector near similar concepts, potentially missing the exact version string that BM25 matches precisely with its bigram tokenizer.

Why BM25 alone fails on semantic queries

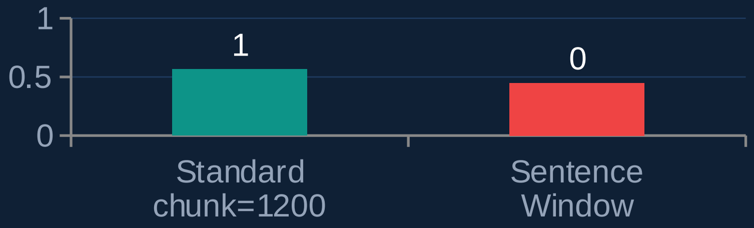
"What tool can I use to design buildings?" — BM25 finds no keyword match unless "design buildings" appears verbatim. Vector search maps this to Revit, AutoCAD Architecture, Forma.

Retrieval — Sentence Window Retrieval (Experiment 06)

Tried, measured, and honestly documented as a negative result

What Is Sentence Window Retrieval?

- Index at sentence granularity (~100–150 chars per chunk)
- At retrieval time: expand each hit to its ±2 surrounding sentence window
- Rationale: precise retrieval units + sufficient context for LLM generation
- Hypothesis: smaller units would improve Context Relevance scores



Why It Underperformed

- 120-char sentence lacks surrounding context for judge LLM to score as 'relevant'
- RAG Triad judge needs enough text to determine topical relevance
- 1,200-char chunk containing the same sentence + explanation scores higher
- Result: CR dropped 0.448 vs 0.564 for standard chunking (corpus mode)

💡 Engineering Takeaway

The right path to higher CR is a stronger reranker or embedding model — not smaller retrieval granularity. Negative results are real results.

Generation — Prompting Strategy & Hallucination Mitigation

Four layered defences against confabulation

LLM Choice

gemma3:4b
~4GB, 30–60 tok/s
Apple Metal GPU
fully local

Why not mistral:7b?
For RAG, the LLM summarises retrieved context. A well-loaded 4B model outperforms a poorly-loaded 7B every time.

Prompting Strategy

01 **Grounded System Prompt**

ONLY answer from provided context. Never use external knowledge.

02 **Hard Refusal Instruction**

If question is off-topic: decline immediately, do not engage with the subject.

03 **Score-Weighted Context**

Chunks sorted by rerank score — LLM sees most reliable sources first.

04 **Format Hints per Category**

Comparison queries → structured answer. Pricing → only state if in context.

Evaluation — TruLens RAG Triad Methodology

LLM-as-judge · post-hoc scoring · 10 experiments tracked

Context Relevance

Avg: ~0.56

Are retrieved chunks relevant to the query?

Low = retrieval failure — wrong chunks surfaced

Groundedness

Avg: ~0.97

Is every claim in the answer supported by retrieved context?

Low = hallucination

Answer Relevance

Avg: ~0.75

Does the answer address what was actually asked?

Low = off-topic

Composite = geometric mean A single weak leg collapses the composite — all three must be strong simultaneously. *Methodology traceable to RAGAS (Es et al., 2023)*

Evaluation — Architecture & Test Suite Design

Post-hoc scoring · decoupled test suite · two-app dashboard

Post-Hoc Scoring Pattern

① All RAG generation completes first

② Judge LLM scores in a second pass

③ Scores batch-written to TruLens DB

④ JSON export to experiment_results/

→ Judge latency never on the user-facing critical path
→ Context capture fix: cache full RAGResponse so web chunks (blended) are visible to scorer

Test Suite: 9 Categories, 15 Questions

product_info
General product knowledge

comparison
Multi-document synthesis

feature_query
Specific capability lookup

pricing_adversarial
Hallucination probe — no price in corpus

version_adversarial
Must not invent version numbers

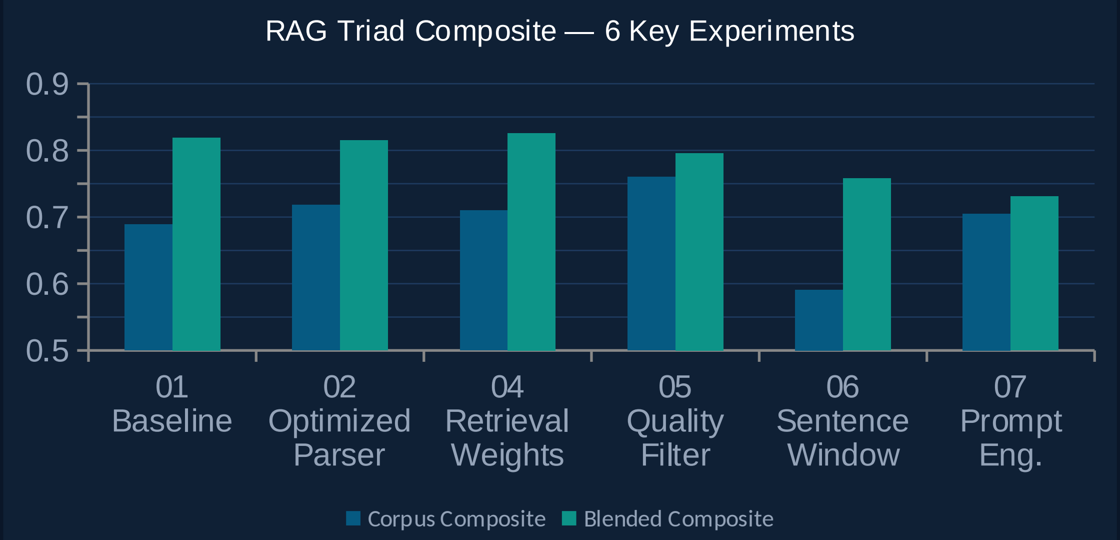
recency_probe
Corpus gap — blended mode advantage

irrelevant
Graceful refusal required

Test suite decoupled to JSON — non-engineers can add questions without touching Python

Insights & Edge Cases — Metric Results

What the numbers reveal and what they hide



Edge Case Analysis

irrelevant category

CR = 0.04–0.10 (correct behaviour, not failure). Groundedness = ~0.9 (refusals are vacuously grounded). Removing from aggregate shifts CR from 0.56 → 0.63.

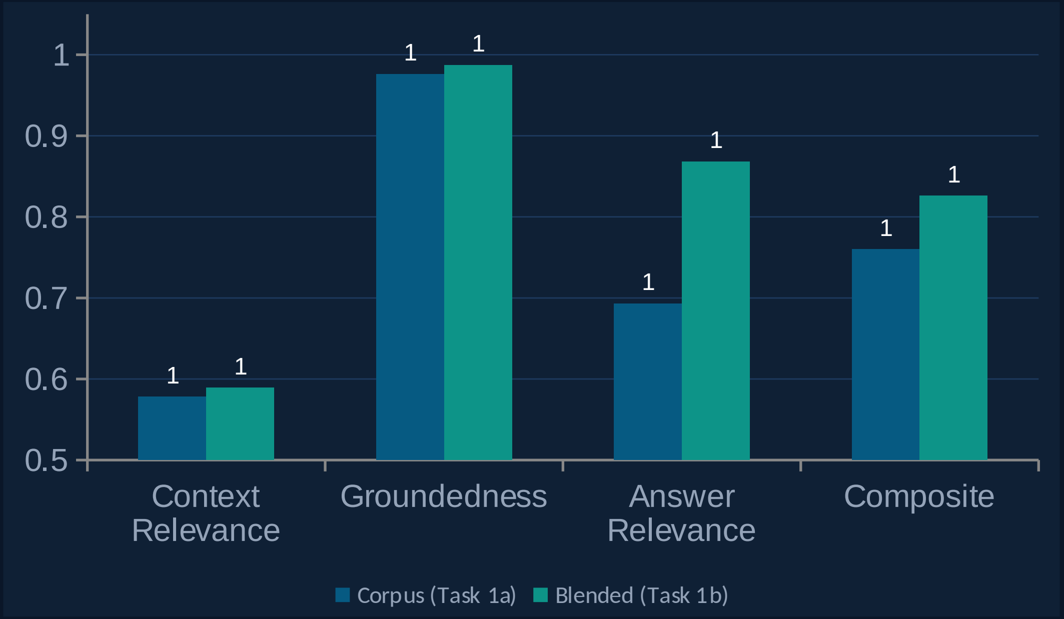
recency_probe category

CR = 0.07–0.20 corpus, 0.20–0.46 blended. Not a retrieval algorithm failure — the corpus was crawled before 2025 AI features were published. Static corpus limitation, not a model failure.

Key finding: Context Relevance (avg ~0.56) is the persistent weak leg. Groundedness (avg ~0.97) confirms grounded prompting is working. The CR floor is partly structural — irrelevant and recency categories inflate the deficit. Removing them gives a fairer corpus CR of ~0.65.

Task 1a vs Task 1b — Corpus vs Blended Web Search

When does web search help, and when does it hurt?



Answer Relevance: +0.175

Blended significantly better — web fills gaps for recency probes and out-of-corpus questions. Biggest win category.

Groundedness: +0.011

Groundedness held. Web results are contributing grounded, verifiable content rather than introducing noise.

Latency: +3,700ms avg

DuckDuckGo adds ~3.5s per query. Acceptable for quality gain but must be monitored in production.

Context Relevance: +0.011

Minimal difference — web search adds context but the judge scores it similarly. No meaningful degradation.

PoC to Production — 4-Week Project Plan

What a full-featured product would look like

Week 1

Data & Infrastructure

- Playwright crawler for JS-rendered pages
- PostgreSQL: conversations + feedback
- ChromaDB → Qdrant (production vector search)
- CI/CD pipeline + automated testing

Week 2

Retrieval Quality

- bge-large-en-v1.5 (~5 CR points gain)
- MiniLM-L-12 reranker (no re-ingest)
- Query understanding: intent + entity extraction
- Auto-merging retrieval for comparison queries

Week 3

Agent & Generation

- LangGraph multi-agent: Router → Retrieval → Web
- Tool-use: pricing API, product catalog
- GPT-4o judge (eliminate same-model bias)
- RLHF from collected thumbs up/down signals

Week 4

Deploy & Monitor

- Kubernetes deployment + auto-scaling
- Latency P99, error rate, hallucination tracking
- CI evaluation gate on every config change
- User analytics: popular queries, failure patterns

✅ Already implemented: LLM-as-judge RAG Triad evaluation, hybrid retrieval, blended web search, conversation history, human feedback, Docker

Reflections & What I'd Do With More Time

Honest assessment of choices, trade-offs, and next highest-impact improvements

01

Data quality is the #1 lever

Better HTML parsing → better chunks → better retrieval → better answers. The v1 precision-mode bug silently zeroed recall for competitor comparisons. The iteration loop (explore → measure → fix) is the core engineering value here.

02

Hybrid search is non-negotiable for product corpora

"AutoCAD LT 2024" — BM25 catches exact version strings that dense vectors miss. The bigram tokenizer fix for compound product names was a high-ROI change with immediate, measurable impact.

03

Evaluation validity requires honest framing

CR avg ~0.56 looks weak. But irrelevant (correct refusal) and recency (corpus gap) categories structurally deflate it. Presenting raw and category-adjusted numbers tells the complete story.

04

Negative results are real results

Sentence Window Retrieval (Exp 06) lowered CR vs standard chunking. Documenting why — retrieval granularity vs judge context needs — is more valuable than hiding the result.

Next highest-impact improvements: bge-large-en-v1.5 embeddings (est. +5 CR points) · MiniLM-L-12 reranker (no re-ingest) · GPT-4o

Thank You

Questions & Discussion

10 Experiments Tracked	**3** Retrieval Strategies	**15** Eval Questions	**0** External API Keys

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